Arc-Sim -- Weather-Driven Design Report

Generated: 2026-09-03 21:20:36
Run folder: CSWARCMOD1E_weatherdesign_20260903_153133

Weather Used

Location: Silver Springs
Fetched: 2026-09-03 15:31:12
Wind avg / gust: 4.61 / 7.00 m/s
Direction: 302 deg
Temp / Pressure: 29.3 C / 1012.0 mbar
Condition: Heavy rain at times

Targets

Apogee: 243.84 m
Flight time: 37.5 - 39.5 s

Main Solved Design

Ballast: 87.4 g
Fin height: 0.0385 m
Fin sweep: 0.0550 m (unchanged)
Hole radius: 3.23 in
Apogee: 243.85 m (within tolerance)
Flight time: 39.01 s (within tolerance)

Margin Fin Sets (day-of wind variability, ballast/hole radius held fixed)

wind m/s	fin height m	apogee m	time s
3.66	0.0451	242.80	38.79
4.13	0.0415	242.44	38.79
5.09	0.0394	244.71	39.14
5.57	0.0392	240.72	38.64

Notes

Always verify the stability margin (1-2 calibers) of the solved design in OpenRocket directly before committing to it -- this solver trims fin height/sweep, which can move the CP.
Local-conditions sweep results (day-of variability) are in the companion .xlsx in the run folder.